

Notation.

Let $X: U \rightarrow \mathbb{R}^3$ be a Patch. Define

$$g_{i\dot{j}} \stackrel{\text{def}}{=} \langle X_i, X_{\dot{j}} \rangle$$

The first-fundamental form I on M is

$$I: T_p M \times T_p M \rightarrow \mathbb{R}: (A, B) \mapsto \sum_{1 \leq i, \dot{j} \leq 2} g_{i\dot{j}} \cdot A^i \cdot B^{\dot{j}}$$

And, $g \stackrel{\text{def}}{=} \det(g_{i\dot{j}})$ and $g^{kl} = (k, l)$ entry of $(g_{i\dot{j}})^{-1}$.

The normal of X at $X(s)$ is ($s = (u, v)$)

$$n(s) = \frac{X_1(s) \times X_2(s)}{\|X_1(s) \times X_2(s)\|}.$$

And, the second-derivatives at $X(s)$ is

$$X_{i\dot{j}}(s) = \frac{\partial^2 X}{\partial u^{\dot{i}} \partial v^j}(s)$$

And, the intrinsic normal of a unit speed curve in the image of X is

$$S = n \times T.$$

The Normal and Geodesic Curvature of a unit speed curve γ in X

$$K_n = \langle \gamma''(s), n(\gamma(s), \gamma'(s)) \rangle \quad \text{and} \quad K_g = \langle \gamma''(s), S(s) \rangle$$

The coefficients of the second fundamental form of X is

$$L_{i\dot{j}} = \langle X_{i\dot{j}}, n \rangle \quad (1 \leq i, \dot{j} \leq 2)$$

The second-fundamental form II on M is the bilinear form

$$II: T_p M \times T_p M \rightarrow \mathbb{R}: (X, Y) \mapsto \sum_{1 \leq i, \dot{j} \leq 2} L_{i\dot{j}} \cdot X^i \cdot Y^{\dot{j}}$$

with associated Matrix L . ($L_k^l = \sum L_{i\dot{k}} g^{\dot{i}l}$)

The Christoffel symbols

$$\Gamma_{i\dot{j}}^k \stackrel{\text{def}}{=} \sum_{l=1}^2 \langle X_{i\dot{j}}, X_l \rangle g^{lk}.$$

Debris

So that given two tangent vectors $A = \sum a^i X_i$ and $B = \sum b^{\dot{j}} X_{\dot{j}}$ exists

$$\langle A, B \rangle = \sum_{1 \leq i, \dot{j} \leq 2} a^i b^{\dot{j}} \cdot \langle X_i, X_{\dot{j}} \rangle = \begin{pmatrix} b^1 & b^2 \end{pmatrix} \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{pmatrix} a^1 \\ a^2 \end{pmatrix}$$

Forms of the Monge patch.

Let $X: U \rightarrow \mathbb{R}^3: (u^1, u^2) \mapsto (u^1, u^2, f(u^1, u^2))$ where f is a differentiable.

Then,
$$\begin{cases} X_1 = (1, 0, f_1) \\ X_2 = (0, 1, f_2) \\ n = (-f_1, -f_2, 1) / \sqrt{1 + (f_1)^2 + (f_2)^2} \end{cases} \quad \leftarrow \text{It forms the 1st form.}$$

and

$$\begin{cases} X_{11} = (0, 0, f_{11}) \\ X_{12} = (0, 0, f_{12}) = (0, 0, f_{21}) = X_{21} \\ X_{22} = (0, 0, f_{22}) \end{cases}$$

The associated Matrix of the 1st-form is

$$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} = \begin{bmatrix} 1 + (f_1)^2 & f_1 f_2 \\ f_1 f_2 & 1 + (f_2)^2 \end{bmatrix} \quad (\text{so that } I(A, B) = B^t \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \cdot A)$$

The associated Matrix of the 2nd-form is

$$\begin{bmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{bmatrix} = \frac{1}{\sqrt{1 + (f_1)^2 + (f_2)^2}} \cdot \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix} \quad \leftarrow \text{hess}(f).$$

And the Christoffel symbols is

$$\Gamma_{ij}^k = \frac{f_{ij} \cdot f_k}{1 + (f_1)^2 + (f_2)^2}$$

Full calculation of the Christoffel symbol.

$$\begin{aligned}
 \Gamma_{11}^1 &= \langle X_{11}, X_1 \rangle \cdot g^{11} + \langle X_{11}, X_2 \rangle \cdot g^{21} \\
 &= \langle X_{11}, X_1 \rangle \cdot g_{22} (\det g_{ij})^{-1} + \langle X_{11}, X_2 \rangle \cdot (-g_{21}) \cdot (\det g_{ij})^{-1} \\
 &= (f_{11} \cdot f_1 \cdot [1 + (f_2)^2] + f_{11} \cdot f_2 \cdot (-f_1 \cdot f_2)) / (1 + (f_1)^2 + (f_2)^2 + (f_1 \cdot f_2)^2 - (f_1 \cdot f_2)^2) \\
 &= f_{11} \cdot f_1 / (1 + (f_1)^2 + (f_2)^2)
 \end{aligned}$$

$$\begin{aligned}
 \Gamma_{11}^2 &= \langle X_{11}, X_1 \rangle \cdot g^{12} + \langle X_{11}, X_2 \rangle \cdot g^{22} \\
 &= \langle X_{11}, X_1 \rangle \cdot (-g_{12}) \cdot (\det g_{ij})^{-1} + \langle X_{11}, X_2 \rangle \cdot g_{11} \cdot (\det g_{ij})^{-1} \\
 &= (f_{11} \cdot f_1 \cdot (-f_1 \cdot f_2) + (f_{11} \cdot f_2 \cdot [1 + (f_1)^2])) / (1 + (f_1)^2 + (f_2)^2) \\
 &= f_{11} \cdot f_2 / (1 + (f_1)^2 + (f_2)^2)
 \end{aligned}$$

$$\begin{aligned}
 \Gamma_{12}^1 &= \langle X_{12}, X_1 \rangle \cdot g^{11} + \langle X_{12}, X_2 \rangle \cdot g^{21} \\
 &= \langle X_{12}, X_1 \rangle \cdot g_{22} (\det g_{ij})^{-1} + \langle X_{12}, X_2 \rangle \cdot (-g_{21}) \cdot (\det g_{ij})^{-1} \\
 &= (f_{12} \cdot f_1 \cdot [1 + (f_2)^2] + f_{12} \cdot f_2 \cdot (-f_1 \cdot f_2)) / (1 + (f_1)^2 + (f_2)^2) \\
 &= f_{12} \cdot f_1 / (1 + (f_1)^2 + (f_2)^2)
 \end{aligned}$$

$$\begin{aligned}
 \Gamma_{12}^2 &= \langle X_{12}, X_1 \rangle \cdot g^{12} + \langle X_{12}, X_2 \rangle \cdot g^{22} \\
 &= (f_{12} \cdot f_1 \cdot (-f_1 \cdot f_2) + f_{12} \cdot f_2 \cdot [1 + (f_1)^2]) / (1 + (f_1)^2 + (f_2)^2) \\
 &= (f_{12} \cdot [f_2 + (f_1)^2 \cdot f_2 - (f_1)^2 \cdot f_2]) / (1 + (f_1)^2 + (f_2)^2) \\
 &= f_{12} \cdot f_2 / (1 + (f_1)^2 + (f_2)^2)
 \end{aligned}$$

$$\begin{aligned}
 \Gamma_{22}^1 &= \langle X_{22}, X_1 \rangle \cdot g^{11} + \langle X_{22}, X_2 \rangle \cdot g^{21} \\
 &= (f_{22} \cdot f_1 \cdot [1 + (f_2)^2] + f_{22} \cdot f_2 \cdot (-f_1 \cdot f_2)) / (1 + (f_1)^2 + (f_2)^2) \\
 &= f_{22} \cdot f_1 / (1 + (f_1)^2 + (f_2)^2)
 \end{aligned}$$

$$\Gamma_{22}^2 = f_{22} \cdot f_2 / (1 + (f_1)^2 + (f_2)^2)$$

Shape Operator.

Def. The map

$$L: T_p M \rightarrow \mathbb{R}^3: X \mapsto -Xn$$

(where $n: D(CM) \rightarrow S^2: g \mapsto n(g)$ and Xf is a notation of directional derivative).
is called the shape operator. In actually, $L \in \text{End}(T_p M)$.

Lemma. Let M be a surface and $L: T_p M \rightarrow \mathbb{R}^3: X \mapsto -Xn$ be the shape operator.

$$\begin{aligned} L(a \cdot X_1 + b \cdot X_2) &= \begin{bmatrix} L_{11} \cdot g^{11} + L_{21} \cdot g^{21} & L_{12} \cdot g^{11} + L_{22} \cdot g^{21} \\ L_{11} \cdot g^{12} + L_{21} \cdot g^{22} & L_{12} \cdot g^{12} + L_{22} \cdot g^{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \\ &= \begin{bmatrix} L_1^1 & L_2^1 \\ L_1^2 & L_2^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \text{ where } L_k^l = L_{1k} \cdot g^{1l} + L_{2k} \cdot g^{2l} \end{aligned}$$